

Django Complete Course (Basic to Advanced)

Module 1: Introduction to Django

Lesson 1

- What is Django?
- History of Django
- Why Django?
- Features of Django
- Advantages & Disadvantages
- Django Architecture (MVT)
- Django vs Flask
- Django vs FastAPI
- Installing Django

Commands

```
pip install django  
django-admin --version
```

Module 2: Environment Setup

Lesson 2

- Install Python
- Create Virtual Environment
- Activate Virtual Environment
- Install Django
- Create First Project
- Run Development Server

Commands

```
python -m venv venv
```

```
# Windows
venv\Scripts\activate
```

```
# macOS/Linux
source venv/bin/activate
```

```
pip install django
```

```
django-admin startproject myproject
```

```
cd myproject
```

```
python manage.py runserver
```

Project Structure

```
myproject/
|
|— manage.py
|— myproject/
|   |— settings.py
|   |— urls.py
|   |— asgi.py
|   |— wsgi.py
|   |— __init__.py
```

Module 3: Django Apps

Lesson 3

- What is an App?
- Project vs App
- Create App
- Register App

Commands

```
python manage.py startapp students
```

Projects

- Student App
 - Employee App
-

Module 4: URL Routing

Lesson 4

Topics

- URL Dispatcher
- path()
- include()
- URL Parameters
- Dynamic URLs

Projects

- Student Profile
 - Product Details
-

Module 5: Views

Lesson 5

- Function-Based Views (FBV)
- Returning Responses
- Rendering Templates
- Redirecting Pages

Projects

- Home Page
 - About Page
 - Contact Page
-

Module 6: Templates

Lesson 6

- Template Engine

- Template Tags
- Variables
- Filters
- Template Inheritance
- Static Files

Projects

- Company Website Layout
-

Module 7: Models

Lesson 7

Topics

- Models
- Fields
- Relationships
- Meta Class

Common Fields

- CharField
- IntegerField
- FloatField
- DateField
- DateTimeField
- BooleanField
- ImageField
- FileField
- EmailField

Projects

- Student Model
 - Employee Model
-

Module 8: Migrations

Lesson 8

Commands

`python manage.py makemigrations`

`python manage.py migrate`

`python manage.py showmigrations`

Topics

- Migration Files
 - Database Synchronization
-

Module 9: Django ORM

Lesson 9

Topics

- Create
- Read
- Update
- Delete

Queries

- `filter()`
- `exclude()`
- `get()`
- `order_by()`
- `values()`
- `aggregate()`

Projects

- Student CRUD
-

Module 10: Django Admin

Lesson 10

Topics

- Admin Panel
- Register Models
- ModelAdmin
- List Display
- Search
- Filters

Commands

`python manage.py createsuperuser`

Projects

- Student Admin Panel
-

Module 11: Forms

Lesson 11

Topics

- HTML Forms
- Django Forms
- Model Forms
- Form Validation

Projects

- Student Registration
-

Module 12: Static Files

Lesson 12

Topics

- CSS
- JavaScript
- Images
- Bootstrap Integration

Projects

- Portfolio Website
-

Module 13: Media Files

Lesson 13

Topics

- Image Upload
- File Upload
- MEDIA_ROOT
- MEDIA_URL

Projects

- Student Profile Photo
-

Module 14: Authentication

Lesson 14

Topics

- Register
- Login
- Logout
- Password Reset
- Change Password

Projects

- User Authentication System
-

Module 15: Authorization

Lesson 15

Topics

- Permissions
- Groups
- Decorators
- Login Required

Projects

- Employee Portal
-

Module 16: Class-Based Views

Lesson 16

Topics

- View
- TemplateView
- ListView
- DetailView
- CreateView
- UpdateView
- DeleteView

Projects

- Blog CRUD
-

Module 17: CRUD Operations

Lesson 17

Projects

- Student Management System
 - Employee Management System
-

Module 18: Pagination & Search

Lesson 18

Topics

- Pagination
- Search
- Filtering
- Ordering

Projects

- Product Catalog
-

Module 19: Signals

Lesson 19

Topics

- post_save
- pre_save
- post_delete

Projects

- User Profile Auto Creation
-

Module 20: Email Integration

Lesson 20

Topics

- Gmail SMTP
- OTP
- Contact Form
- Password Reset Email

Projects

- Contact Us Form
-

Module 21: Django REST Framework

Lesson 21

Topics

- DRF Installation
- Serializers
- API Views
- ViewSets
- Routers

Projects

- Student REST API
-

Module 22: JWT Authentication

Lesson 22

Topics

- Token Authentication
- JWT
- Refresh Tokens

Projects

- Secure Login API
-

Module 23: API Integration

Lesson 23

Topics

- Consume External APIs
- JSON Response
- Requests Library

Projects

- Weather Dashboard
 - News App
-

Module 24: Testing

Lesson 24

Topics

- Unit Testing
- Django TestCase
- Client Testing

Projects

- Login Testing
-

Module 25: Caching

Lesson 25

Topics

- Local Memory Cache
- Redis (Introduction)
- Cache Framework

Projects

- Blog Performance
-

Module 26: Security

Lesson 26

Topics

- CSRF
- XSS
- SQL Injection
- Clickjacking
- Secure Cookies
- HTTPS

Projects

- Secure Banking Portal
-

Module 27: Deployment

Lesson 27

Deploy To

- PythonAnywhere
- Render
- Railway
- VPS
- Nginx + Gunicorn

Topics

- Environment Variables
- Static File Collection

Commands

```
python manage.py collectstatic
```

Module 28: Best Practices

Lesson 28

Topics

- Project Structure
 - Reusable Apps
 - Environment Files
 - Logging
 - Code Organization
-

Module 29: Performance Optimization

Lesson 29

Topics

- Query Optimization
- `select_related()`
- `prefetch_related()`
- Indexing
- Lazy Loading

Projects

- High Performance Dashboard
-

Module 30: Final Projects

Build Complete Full-Stack Django Applications

Projects

1. Student Management System
 2. Employee Management System
 3. Library Management System
 4. Hospital Management System
 5. School ERP
 6. College ERP
 7. Banking System
 8. Hotel Booking System
 9. Learning Management System (LMS)
 10. E-Commerce Website
 11. Online Examination System
 12. CRM System
 13. Inventory Management System
 14. Job Portal
 15. Food Delivery Application
 16. Courier Tracking System
 17. Expense Tracker
 18. Blog Management System
 19. Attendance Management System
 20. AI-Based Student Performance Prediction System
-

30 Hands-on Practice Projects

1. Hello Django
2. Student App
3. Employee App
4. URL Routing Demo
5. Template Rendering
6. Student Model
7. Migrations Practice
8. Student CRUD
9. Admin Panel Customization
10. Registration Form
11. Login System
12. Image Upload
13. Blog CRUD
14. Pagination
15. Search Functionality
16. Contact Form
17. Email Integration
18. Student REST API
19. JWT Login API
20. Weather Dashboard
21. Notes Application

22. Inventory System
 23. Hospital Dashboard
 24. School ERP
 25. Hotel Booking System
 26. LMS Portal
 27. E-Commerce Website
 28. Online Exam System
 29. Banking Portal
 30. Complete Company ERP
-

Django Interview Questions (Basic to Advanced)

Beginner

1. What is Django?
2. Explain the MVT architecture.
3. What is a Django project?
4. What is a Django app?
5. What is `manage.py`?
6. What are migrations?
7. What is the ORM?
8. What are models?
9. What are views?
10. What are templates?

Intermediate

11. What are Class-Based Views (CBVs)?
12. Difference between FBVs and CBVs?
13. What are Django Forms?
14. Explain ModelForms.
15. What is the Admin Panel?
16. Explain middleware.
17. What are static and media files?
18. What is Django REST Framework?
19. What are serializers?
20. What is JWT authentication?

Advanced

21. Explain Django Signals.
22. What is QuerySet optimization?

23. Difference between `select_related()` and `prefetch_related()`?
 24. Explain Django caching.
 25. What is CSRF protection?
 26. How do you deploy a Django application?
 27. Explain Django middleware in detail.
 28. How do you scale a Django application?
 29. What are Django best practices?
 30. How do you secure a Django application?
-

20 Real-World Django Projects

1. Student Management System
 2. Employee Management System
 3. Library Management System
 4. Hospital Management System
 5. School ERP
 6. College ERP
 7. Learning Management System (LMS)
 8. E-Commerce Website
 9. Hotel Booking System
 10. Banking Portal
 11. CRM System
 12. Online Examination System
 13. Job Portal
 14. Food Delivery System
 15. Courier Tracking System
 16. Attendance Management System
 17. Blog CMS
 18. Inventory Management System
 19. Expense Tracker
 20. AI-Based Student Performance Prediction System
-

Django Cheat Sheet

Installation

```
pip install django
django-admin --version
```

Project

```
django-admin startproject myproject
```



```
cd myproject
python manage.py runserver
```

App

```
python manage.py startapp students
```

Migrations

```
python manage.py makemigrations
python manage.py migrate
python manage.py createsuperuser
```

Common Commands

```
python manage.py shell
python manage.py collectstatic
python manage.py test
python manage.py runserver
```

Install Packages

```
pip install djangorestframework
pip install djangorestframework-simplejwt
pip install pillow
pip install python-dotenv
pip install mysqlclient
```

Learning Path After Django

1. Python Fundamentals 
2. Object-Oriented Programming 
3. SQL & Database Design
4. Django Fundamentals 
5. Django ORM
6. Authentication & Authorization
7. Django REST Framework (DRF)
8. JWT Authentication
9. React.js Integration
10. Docker
11. Redis
12. Celery
13. Nginx + Gunicorn
14. AWS / Azure Deployment
15. CI/CD Pipelines
16. Microservices
17. GraphQL

18. AI Integration with Django
19. Production Deployment
20. Full Stack Django + React Development

This roadmap is designed for **industrial training programs, LMS portals, and job-oriented full-stack web development**, taking learners from Django fundamentals to advanced backend development, REST APIs, authentication, deployment, performance optimization, security, and enterprise-level application development.